

■ ATS Resume Parser — Project Progress Report

Comprehensive Report of Development, Architecture, and Features Implemented

■ Executive Summary

The **ATS Resume Parser** has evolved from a basic Streamlit script into a full-stack, decoupled, production-ready AI application. The system extracts unstructured resume content (text + embedded PDF hyperlinks) and uses **Groq-accelerated LLMs** (openai/gpt-oss-120b) to convert candidates' resumes into standardized, structured JSON profiles.

■■ Architectural Evolution

Phase 1: Legacy Monolithic Prototype

PDF Upload → Streamlit UI (app.py) → PyPDF Text Extraction → Groq API LLM Call

Phase 2: Modern Decoupled Architecture (Current)

React + Vite + Tailwind UI → FastAPI Backend (api.py) → PyPDF (Text & Links) → Groq LLM Engine → Candidate Profile Dashboard

■ Features Implemented

1. Hyperlink & Text Extraction Engine (PyPDF)

- **Visible Text:** Reads page-by-page unstructured text from uploaded PDFs.
- **Embedded Link Extraction:** Parses PDF annotation objects (/Annots → /URI) to capture hidden LinkedIn, GitHub, and Portfolio URLs.
- **Consolidated Input:** Appends extracted URLs to raw text prompt ensuring no links are omitted.

2. Groq LLM Parsing Engine (resumeparser.py)

- **Structured JSON Schema:** Enforces strict schema using `response_format={'type': 'json_object'}`.
- **LLM Model:** Powered by openai/gpt-oss-120b for fast, accurate extraction.
- **Deterministic Output:** Set temperature to 0.0 and expanded token budget to 4000 max tokens.
- **Fields Extracted:** Full Name, Email, Phone, Social/Portfolio URLs, Technical Skills, Soft Skills, Education, Employment, Projects, and Achievements.

3. FastAPI Backend API (api.py)

- **Endpoint:** POST /api/parse-resume handling PDF upload via multipart form-data.
- **CORS Middleware:** Enables full client-server decoupling.

- **Validation & Sanitization:** Enforces PDF format checks and auto-cleans markdown JSON code fences.

4. Modern React + Vite + Tailwind Frontend (frontend/)

- **Drag-and-Drop Dropzone:** Interactive uploader with visual feedback and file size metadata.
- **Candidate Dashboard:** Displays profile header, brand-colored social buttons, skills pills, education cards, project grid, and achievements breakdown.
- **Aesthetics:** Built with dark glassmorphic design, slate themes, indigo/emerald highlights, and Lucide React icons.

■■ Tech Stack Breakdown

Layer	Technology	Usage / Description
Frontend	React 18 + Vite	Modern, ultra-fast single-page web app
Styling	Tailwind CSS + Lucide Icons	Responsive glassmorphic dark theme
Backend API	FastAPI + Uvicorn	Async RESTful server with CORS middleware
PDF Processing	PyPDF	Text & embedded URI hyperlink annotation parser
LLM Engine	Groq SDK (openai/gpt-oss-120b)	High-speed deterministic JSON parsing engine
Legacy Interface	Streamlit	Standalone single-file Python prototype UI

■ Commit History & Key Milestones

- c4fb620** - Initial commit: Streamlit prototype in app.py with basic PyPDF reader & Procline.
- 41717be & 6aafed** - Render deployment prep & GROQ_API_KEY environment variable integration.
- 3fc4096** - Repository cleanup: untracked IDE files and dataset directories in .gitignore.
- 94886d4** - Full-stack transformation: introduced FastAPI backend (api.py) and React + Vite frontend.
- 46d0e9f** - Comprehensive documentation: updated README.md with setup guides.
- 0cbe9e1** - Extended LLM schema & UI for Achievements & Honors extraction.

■ Project File Structure

```
resume-parser/  
■■■ api.py           # FastAPI backend server (POST /api/parse-resume)  
■■■ resumeparser.py  # PDF text/link extraction + Groq API prompt logic  
■■■ app.py           # Legacy Streamlit app  
■■■ requirements.txt  # Python dependencies  
■■■ README.md        # Documentation & usage instructions  
■■■ frontend/        # React + Vite application  
    ■■■ src/  
        ■■■ App.jsx   # Main Dashboard & Dropzone Component  
        ■■■ index.css# Tailwind CSS design system
```

```
■■■ package.json # Frontend dependencies
■■■ vite.config.js
```

■ Potential Next Steps

- **Multi-Resume Batch Upload:** Allow bulk uploading and parsing of multiple PDF resumes at once.
- **Export Options:** Download candidate profiles as JSON, CSV, or formatted PDF summaries.
- **ATS Compatibility Scoring:** Match extracted resume skills against job description (JD) prompts.
- **Database Persistence:** Connect SQLite / PostgreSQL to store parsed candidate history.